Aishwarya L Pujeri

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PROFILE

Aspiring Artificial Intelligence Engineer with hands-on experience in developing intelligent applications across Machine Learning, Deep Learning, NLP, and Computer Vision. Proficient in Python with practical experience in Scikit-learn, TensorFlow, PyTorch, YOLO, OpenCV, and NLTK. Experienced in data preprocessing, feature engineering, model development, classification, and performance evaluation. Skilled in building and deploying AI applications using Streamlit, with a focus on solving real-world problems through data-driven and intelligent solutions.

EDUCATION

M. Tech in Artificial Intelligence, *(Full-time)* Nov 2025 – Nov 2027
REVA Academy for Corporate Excellence, REVA University
8.5 CGPA

B.E in Computer Science and Engineering, *(Full-time)* Dec 2021 – Jun 2025
Government Engineering College Ramanagara, VTU
7.51 CGPA

PU in PCMB, *KLE's Angadi PU College Gokak (Full-time)* Jun 2019 – Jun 2021
93%

SSLC, *Modern English Medium School Gokak (Full-time)* Jun 2018 – Jun 2019
92%

SKILLS

Programming Languages
Python, Java, HTML, CSS, SQL

Data Science and Visualization
NumPy, Pandas, Matplotlib, Seaborn, Power BI

Machine Learning
Scikit-learn, Regression, Classification, Clustering, Feature Engineering, Data Preprocessing, EDA, Model Evaluation, Hyperparameter Tuning

Natural Language Processing (NLP)
NLTK, Text Preprocessing, TF-IDF, Word Embeddings

Deep Learning and Computer Vision
CNN, LSTM, RNN, YOLOv5, YOLOv8, Object Detection, Image Classification

Tools & Platforms
VS Code, Jupyter Notebook, MySQL Workbench, Google colab, GitHub, Eclipse IDE, MS Excel, Power BI

Soft Skills
Positive Attitude, Quick Learner, Adaptability, Teamwork, Time Management

INTERNSHIP

Java Full Stack Intern, *Dhee Coding Lab* Feb 2025 – Jun 2025

- Gained hands-on experience in Java, MySQL, HTML, and CSS by developing full-stack web applications, implementing CRUD operations, and integrating front-end interfaces with back-end database functionality.
- Strengthened skills in Object-Oriented Programming (OOP), SQL, database design, debugging, code optimization, and responsive web development through practical assignments and project-based learning.

PROJECTS

Face Recognition Attendance System | FastAPI, InsightFace, React (Vite), SQLite, Tailwind CSS

- Built a real-time AI attendance system using InsightFace for face detection and recognition, achieving multi-face detection with bounding-box overlays and live confidence scoring directly from webcam frames.
- Designed a modular FastAPI backend with SQLite that enforces one-attendance-per-student-per-day via database-level constraints, with automated CSV, Excel, and PDF report generation.
- Developed a full React + Tailwind dashboard with guided webcam-based student registration, live recognition feed, and Recharts-powered analytics for daily, department-wise, and monthly attendance trends.

Driver Drowsiness & Distraction Detection System | Python, FastAPI, MediaPipe, YOLOv8, JavaScript

- Developed a real-time driver drowsiness and distraction detection system using MediaPipe and Python to compute Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) from live webcam video.
- Built an explainable risk-scoring engine using Python and JavaScript that combines drowsiness and distraction signals into a weighted risk score with human-readable reasons for each alert.
- Integrated a FastAPI backend with JWT authentication, SQLAlchemy, and a YOLOv8-based phone detection model to persist sessions and expose real-time analytics through a Chart.js dashboard.

Crop Yeild Prediction and Crop Recommendation System | Python, Pandas, NumPy, Scikit-learn, Matplotlib

- Developed a machine learning-based crop yield prediction and crop recommendation system using agricultural and environmental datasets to support data-driven farming decisions.
- Performed data preprocessing, exploratory data analysis (EDA), feature engineering, and visualization to identify key factors influencing crop yield and crop selection.
- Implemented and compared machine learning algorithms, including Random Forest and XGBoost, using cross-validation, hyperparameter tuning, and performance evaluation metrics to identify the best-performing models.

House Price Prediction | Python, NumPy, PyPI Packaging, Selenium/Scraping, Scikit-learn.

- Architected and published a 10-module NumPy-engineered machine learning framework from scratch on PyPI, writing core mathematics for OLS, Ridge, Lasso, KNN, PCA, and Neural Networks.
- Built an end-to-end valuation system by engineering a web-scraper script to gather 500+ live real estate listings from MagicBricks.
- Implemented a complete house price prediction pipeline using the custom machine learning package and compared its performance with equivalent Scikit-learn models to validate accuracy and consistency.

Heart Disease Prediction | Python, Pandas, scikit-learn, Classification algorithms, Streamlit

- Developed a binary classification system using the Kaggle Cardiovascular Disease Dataset (cardio_train.csv, 70,000 records), performing data cleaning, feature engineering (BMI, pulse pressure, MAP, and BP ratio), preprocessing, and stratified data splitting.
- Implemented and compared 5 classification algorithms — Logistic Regression, Decision Tree, Random Forest, SVM, and KNN using Accuracy, Precision, Recall, F1-score, and ROC-AUC; selected Random Forest based on the highest validation ROC-AUC (0.7966).
- Developed an interactive Streamlit deployment using serialized Joblib classification models, enabling real-time class prediction, estimated probability visualization, patient analytics, model comparison, and downloadable reports.

AI-Based Food Nutrition Analyzer using RAG and LLM| Python, Streamlit, Langchain, Sentence Transformer, Nutrition API

- Developed an AI-based Food Nutrition Analyzer using Python and Streamlit to retrieve and display nutritional information through a nutrition API.
- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain, FAISS, and Sentence Transformers to retrieve relevant nutrition information from a custom PDF knowledge base.
- Integrated a local Llama 3.2 model via Ollama to generate context-aware responses for nutrition-related user queries based on retrieved documents.

Fake News Detection System ML and DL models | Python, Scikit-learn, TensorFlow, NLTK

- Developed a fake news classification system using ML and DL models trained on ISOT and WELFake datasets to distinguish between real and fake news articles.
- Applied NLP techniques including text preprocessing, tokenization, TF-IDF, and embeddings for feature extraction and model input representation.

- Evaluated and compared multiple models (e.g., Logistic Regression, LSTM, CNN) to achieve improved accuracy and robust generalization on unseen news data.

Bus Ticket Booking System | Python, MySQL, Flask

- Designed and developed a full-stack bus ticket booking web application enabling users to search routes, view available buses, and book tickets in real time.
- Implemented backend functionality using Python and MySQL for user management, booking records, seat allocation, and secure data storage.
- Built an interactive and responsive frontend using HTML, CSS, and JavaScript to enhance user experience and streamline the booking process.

Automated Duplicate File Finder | Python, PyQt, Multithreading, SHA-256 Hashing, NLP

- Engineered a high-performance PyQt desktop utility optimized to locate, isolate, and manage duplicate disk storage assets. Implemented a concurrent scanning architecture utilizing Python's ThreadPoolExecutor to prevent UI freezing during heavy IO operations.
- Utilized SHA-256 hashing to maintain 100% precision in byte-level duplicate classification across dense directory paths.
- Embedded an isolated, rule-based NLP chatbot module to interface with users, analyzing file metadata to offer custom storage usage statistics.

Suspicious Activity Detection using DL models | Python, OpenCV, TensorFlow, PyTorch, YOLOv5, Flask.

- Developed a deep learning security pipeline designed to monitor surveillance video streams and trigger automated alerts for abnormal events (fights, weapons, accidents).
- Integrated a pre-trained YOLOv5 network for real-time spatial object bounding box generation.
- Coupled CNN spatial networks with Long Short-Term Memory (LSTM) layers to handle multi-frame temporal video sequence classification, serving the pipeline via a Flask web layout.

CERTIFICATIONS

- Python 101 for Data Science certification by IBM
- SQL and Relational Database 101 from IBM
- Machine Learning with Python by IBM